

Three-Dimensional Near-Field Microwave Holography Using Reflected and Transmitted Signals

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Abstract—A new 3-D holographic microwave imaging technique is proposed to reconstruct targets in the near-field range. It is based on the Fourier analysis of the wideband transmission and reflection signals recorded by two antennas scanning together along two rectangular parallel apertures on both sides of the inspected region. The complex scattering parameters of the two antennas are collected at several frequencies and then processed to obtain a representation of the 3-D target in terms of 2-D slice images at all desired range locations. No assumptions are made about the incident field and Green's function, which are derived either by simulation or by measurement. Furthermore, an approach is proposed to reduce the image artifacts along range. To validate the proposed technique, predetermined simulated targets are reconstructed. The effects of random noise, number of sampling frequencies, and dielectric contrast of the targets are also discussed.

Index Terms—Image reconstruction, microwave holography, near-field microwave imaging.

I. INTRODUCTION

THE relatively long wavelength of microwave and millimeter waves allows for penetration inside dielectric bodies where visible light cannot reach. Various techniques have been proposed to harness the ability of microwaves for 2-D and 3-D imaging in a wide variety of applications such as biomedical imaging [1], concealed weapon detection [2], through-the-wall imaging [3], nondestructive testing and evaluation [4], etc. All these methods operate on the transmitted and/or received signals at the antenna terminals but they differ significantly in the employed algorithms. Microwave imaging techniques in general could be categorized into: optimization-based microwave imaging (e.g., see [5]), confocal radar-based imaging (e.g., see [6]), microwave tomography (e.g., see [7]), and microwave holography [8].

In optimization-based imaging, the exact nonlinear diffraction relation between the measured scattered field and the complex permittivity is found iteratively. The distribution of dielectric properties is sought within the imaged region employing a forward model. These techniques are in principle capable of achieving super-resolution (e.g., 0.1λ in [9]). However, their

high computational cost (e.g., see [10]) precludes fast performance. Also, they suffer from ill-posedness (e.g., see [11]) and their performance depends critically on the fidelity of the forward model. It is often the case that microwave imaging setups are difficult to align well with electromagnetic numerical simulation.

Confocal microwave imaging aims to detect strong scatterers inside the inspected region [6]. A wide-band pulse is transmitted and the reflections from the targets are focused synthetically. The focusing algorithm constructs a map of the scatterers (if present) inside the interrogated region. While very fast, these techniques are sensitive to clutter [12] and the heterogeneity of the inspected region [13].

Microwave tomography includes computed tomography (CT) [14]–[17] and diffraction tomography (DT) [18]–[22]. Both approaches acquire the transmitted signals in multiple planes intersecting the imaged object. The signals acquired in each plane are used to reconstruct a 2-D slice image in this plane only. Thus, the reconstruction is carried out slice by slice, each slice being processed independently. Both approaches exploit linearization approximations to achieve direct (non-iterative) image reconstruction. Microwave CT [14] employs a straight-path assumption for the rays between the transmitter and receiver and applies filtered backprojection (FBP), similar to X-ray CT. As the straight-ray assumption is often inadequate, microwave CT often employs chirp-pulse radar [15]–[17]. Microwave DT employs the linear Born approximation [23] to the solution of the scattering problem which is valid for weak scatterers. This approximation allows for image reconstruction through a Fourier transform (FT). The processing is in quasi-real time but artifacts are often present if the scatterers are not weak. Also, the spatial resolution is limited by the wavelength. Note that similar Fourier-based reconstruction is employed in synthetic aperture radar (SAR) (e.g., see [24], [25]).

The conventional Gabor holography (optical holography) [26], [27] has been extended to microwave frequencies [28], [29] and to acoustic problems [30], [31]. In these approaches, the interference pattern of a reference wave with the wave scattered from the target is recorded on an aperture called hologram. The hologram is illuminated with the reference wave to reconstruct an image of the object. Modern microwave holography [32]–[35], however, is quite distinct from these conventional approaches. Coherent (magnitude and phase) back-scattered signals are acquired on a surface similarly to the conventional holography. However, the reconstruction of the target is based on a sequence of direct and inverse FTs. This reconstruction is similar to DT in the sense that it employs the linear Born approximation and FTs. However, unlike DT, where reconstruction is done slice by slice,

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the data acquired on the surface is used simultaneously in a single reconstruction process to obtain the 3-D reflectivity distribution of the target. Thus, microwave holography provides a framework where, with wideband frequency information, a 3-D image of the target is obtained. In the case of a single-frequency measurement, microwave holography can provide a 2-D image of the target's cross-section in a plane parallel to the acquisition plane (e.g., see [32] and [35]). Microwave holographic imaging based on rectangular and cylindrical aperture scanning has proven reliable and is now widely employed in concealed weapon detection [32]–[34].

Recently, we extended the single-frequency 2-D holographic image reconstruction to near-field microwave imaging [35]. This method employs not only the back-scattered (as in [32], [33]) but also the forward-scattered signals. The additional information from the forward-scattered signals improves the image quality and enables localization of the target in the range direction. In contrast to previous work, this method does not make any assumptions about the incident field such as plane, spherical or cylindrical wave representations. The incident field is derived in a numeric form either through simulation or measurement. This is important in near-field imaging where the target is close to the antenna and the planar or spherical approximations of the illuminating wave are not valid.

Here, we extend the 2-D near-field holographic imaging technique [35] to full 3-D imaging when wideband information is available. The proposed method has a number of distinct features and advantages compared to the existing 3-D holographic techniques. First, the method allows for incorporating forward-scattered signals in addition to the back-scattered signals. This additional information leads to more accurate reconstruction results and also allows for the significant suppression of image artifacts in the range direction. Second, the method allows for an incident-field distribution represented in numeric form. This distribution can be obtained either through simulation or measurement with the particular antenna setup and medium. Third, it also allows for numeric input of the Green's function, i.e., the set of signals due to point scatterers in the given medium and received by the given antennas. These can be efficiently obtained through simulation as explained later. The accurate representations of the incident field and Green's function for the particular problem at hand are crucial in near-field imaging where analytical approximations such as plane or spherical waves are not adequate. Fourth, the numerical formats of the incident field and the Green function necessitate a new inversion procedure. Previous 3-D holography methods [32], [33] rely on the analytical (exponential) form of the incident field and the Green function in order to cast the inversion expression in the form of a 3-D inverse FT. Resampling of the data in k_z -space is also necessary, which may lead to errors. This procedure is inapplicable with numeric representations of the incident field and the Green's function. Instead, we solve a system of equations in each spatial frequency pair (k_x, k_y) and apply 2-D inverse FT to the least-square solution at each desired range location. The systems of equations have much smaller dimensions compared to the systems of equations in regular optimization-based microwave imaging techniques. This reduces the ill-posedness of the proposed imaging technique significantly. Thus, the 3-D target is reconstructed as a set of 2-D slice images

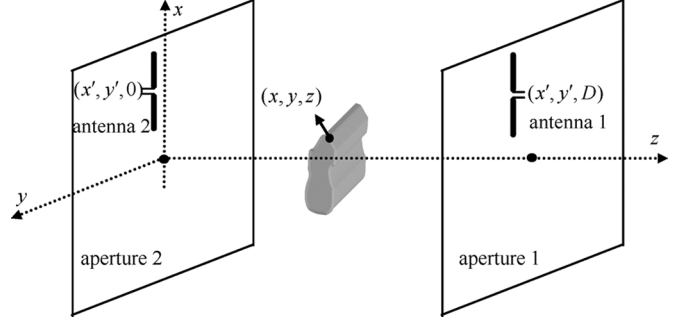


Fig. 1. 3-D Microwave holography setup.

in parallel planes along the range. Fifth, the algorithm is fast, accurate, and robust to high levels of noise. We examine the performance of the 3-D image reconstruction technique by a number of simulation examples.

II. THEORY

The microwave holography set-up in this work consists of two x -polarized antennas and a 3-D target in between as shown in Fig. 1. The formulation of the vectorial scattering problem is presented in the Appendix. Here, the radiation field of the x -polarized antennas (dipoles) can be reasonably approximated by TM_x polarization. Thus, we consider the x -components of the incident and scattered fields only. This leads to a scalar Green's function which is the G_x^x element of the full dyadic. The antennas perform a 2-D scan while moving together on two separate parallel planes positioned at $z = 0$ and $z = D$.

Assume that at any measurement frequency f_l ($l = 1, 2, \dots, N_f$), we know the incident field $E_x^{\text{inc}}(0, 0, 0; x, y, z; f_l)$ at any point (x, y, z) in the inspected volume when the transmitting antenna is at $(0, 0, 0)$. In addition, the Green's function $G_x^x(x, y, z; 0, 0, D; f_l)$ is known for an x -polarized point source at (x, y, z) and E_x measured at $(0, 0, D)$. This information is obtained via simulations as explained later. For brevity, we set

$$E_x^{\text{inc}}(x, y, z, f_l) \equiv E_x^{\text{inc}}(0, 0, 0; x, y, z; f_l) \quad (1)$$

$$G_x^x(x, y, z, f_l) \equiv G_x^x(x, y, z; 0, 0, D; f_l). \quad (2)$$

Let the signal $t(x', y', f_l)$ be the scattered wave received at (x', y', D) , i.e., $t(x', y', f_l) = E_x^{\text{sc}}(x', y', D, f_l)$ where the scattered field is calculated as in (A.8). Note that this implies that the transmitting antenna is at $(x', y', 0)$ since it moves together with the receiving antenna. The incident field and the Green functions for the case where the antenna pair is at (x', y') can be obtained from those in (1) and (2) by a simple translation:

$$E_x^{\text{inc}}(x', y', 0; x, y, z; f_l) = E_x^{\text{inc}}(x - x', y - y', z, f_l) \quad (3)$$

$$G_x^x(x', y', 0; x, y, z; f_l) = G_x^x(x - x', y - y', z, f_l). \quad (4)$$

According to (A.8), $t(x', y', f_l)$ is expressed as

$$t(x', y', f_l) = \int_z \int_y \int_x [k_s^2(x, y, z, f_l) - k_b^2(f_l)] \times E_x^{\text{inc}}(x - x', y - y', z, f_l) \cdot G_x^x(x - x', y - y', z, f_l) dx dy dz. \quad (5)$$

Equation (5) can then be rewritten as

$$t(x', y', f_l) = \int_z \int_y \int_x f(x, y, z, f_l) \cdot g_0(x' - x, y' - y, z, f_l) dx dy dz \quad (6)$$

where

$$f(x, y, z, f_l) = k_s^2(x, y, z, f_l) - k_b^2(f_l) \quad (7)$$

$$g_0(x, y, z, f_l) = E_x^{\text{inc}}(-x, -y, z, f_l) G_x^x(-x, -y, z, f_l). \quad (8)$$

We refer to $f(x, y, z, f_l)$ as the contrast function. Here, the integral over x and y can be interpreted as a 2-D convolution integral. Thus, the 2-D FT of $t(x', y', f_l)$ is written as

$$T(k_x, k_y, f_l) = \int_z F(k_x, k_y, z, f_l) G_0(k_x, k_y, z, f_l) dz \quad (9)$$

where $F(k_x, k_y, z, f_l)$ and $G_0(k_x, k_y, z, f_l)$ are the 2-D FTs of $f(x, y, z, f_l)$ and $g_0(x, y, z, f_l)$, respectively; k_x and k_y are the Fourier variables with respect to x and y , respectively.

To reconstruct the contrast function, we first approximate the integral in (9) by a discrete sum with respect to z for N_z reconstruction planes

$$T(k_x, k_y, f_l) = \sum_{n=1}^{N_z} F(k_x, k_y, z_n, f_l) \times G_0(k_x, k_y, z_n, f_l) \Delta z \quad (10)$$

where Δz is the distance between two neighboring reconstruction planes.

Since we perform measurements at N_f frequencies, writing (10) for all frequencies leads to N_f equations at each spatial frequency pair (k_x, k_y) , see (11).

$$\begin{aligned} T(k_x, k_y, f_1) &= G_0(k_x, k_y, z_1, f_1) F(k_x, k_y, z_1, f_1) + \dots \\ &\quad + G_0(k_x, k_y, z_{N_z}, f_1) F(k_x, k_y, z_{N_z}, f_1) \\ &\vdots \\ T(k_x, k_y, f_{N_f}) &= G_0(k_x, k_y, z_1, f_{N_f}) F(k_x, k_y, z_1, f_{N_f}) + \dots \\ &\quad + G_0(k_x, k_y, z_{N_z}, f_{N_f}) F(k_x, k_y, z_{N_z}, f_{N_f}). \end{aligned} \quad (11)$$

In (11), we have N_f decoupled equations from which finding directly the frequency dependant function $F(k_x, k_y, z_n, f_l)$, $n = 1, 2, \dots, N_z$ and $l = 1, 2, \dots, N_f$, is not feasible. However, under certain assumptions, a system of equations can be constructed from which the contrast function is found.

First, we assume that the contrast function can be expressed as

$$f(x, y, z, f) = h(x, y, z) \phi(f) \quad (12)$$

where the variation of the contrast function with space and frequency is separated into the functions $h(x, y, z)$ and $\phi(f)$,

respectively. Second, we assume that the variation of the contrast function with frequency $\phi(f)$ is known *a priori*. This function can be approximated for example by fitting Cole-Cole or Debye models (e.g., see [36]) to the dielectric properties of the background medium. Then, $\phi(f)$ would approximate the frequency behavior of both the background medium and the targets. In many applications where the operating frequency band is narrow (or in nondispersive mediums), we assume $\phi(f) = 1$. This is the case for the examples presented in this paper. Taking the 2-D FT of both sides of (11) with respect to x and y leads to $F(k_x, k_y, z, f) = H(k_x, k_y, z) \phi(f)$ where $H(k_x, k_y, z)$ is the 2-D FT of $h(x, y, z)$. Thus, replacing $F(k_x, k_y, z, f)$ with $H(k_x, k_y, z) \phi(f)$ in (11) leads to the following system of equations:

$$\begin{cases} y_1 = a_{11}x_1 + \dots + a_{1N_z}x_{N_z} \\ \vdots \\ y_{N_f} = a_{N_f1}x_1 + \dots + a_{N_fN_z}x_{N_z} \end{cases} \quad (13)$$

where

$$y_l = T(k_x, k_y, f_l) \quad (14)$$

$$a_{ln} = G_0(k_x, k_y, z_n, f_l) \phi(f_l) \quad (15)$$

$$x_n = H(k_x, k_y, z_n). \quad (16)$$

This system of equations is solved in the least-square sense to find $H(k_x, k_y, z_n)$, $n = 1, 2, \dots, N_z$, at each spatial frequency pair (k_x, k_y) . Then, inverse 2-D FT is applied to $H(k_x, k_y, z_n)$, $n = 1, 2, \dots, N_z$, to reconstruct a 2-D slice of the function $h(x, y, z_n)$ at each $z = z_n$ plane. Then, the normalized modulus of $h(x, y, z_n)$, $|h(x, y, z_n)|/M$, where M is the maximum of $|h(x, y, z_n)|$ for all z_n , is plotted versus the spatial coordinates x and y to obtain a 2-D image of the target at each $z = z_n$ plane. By putting together all 2-D slice images, a 3-D image of the target is obtained.

For the setup shown in Fig. 1, the acquired scattered waves are obtained by measuring the four complex S -parameters at the two antenna terminals at each scanning position on the apertures. These four S -parameters constitute four separate scattered signals $t(x', y', f_l)$ expressed in (5), from each one a complete set of 2-D slice images can be obtained. The S -parameter s_{jk} ($j, k = 1, 2$) is the scattered signal at the terminals of the j th antenna when the k th antenna transmits. The parameters s_{11} and s_{22} are referred to as the reflection S -parameters while s_{21} and s_{12} are the transmission S -parameters. s_{11} , s_{21} , s_{12} , s_{22} are referred to as the reflection S -parameters while s_{21} , s_{12} , and S_{22} .

In the proposed reconstruction process, we assumed that at any measurement frequency f_l ($l = 1, 2, \dots, N_f$), we know the incident field $E_x^{\text{inc}}(x, y, z, f_l)$ and the Green function $G_x^x(x, y, z, f_l)$ at any target point (x, y, z) when the transmitting and the receiving antennas are at $(0, 0, 0)$ and $(0, 0, D)$, respectively. In prior work exploiting microwave holography, these are approximated as $E_x^{\text{inc}}(x, y, z, f_l) \approx G_x^x(x, y, z, f_l) \approx e^{-jkr}$ which is suitable for far-field scattering problems.

Here, these functions are obtained via simulation of the case when the targets are absent. To obtain $E_x^{\text{inc}}(x, y, z, f)$, we place the transmitting antenna at the origin of the corresponding aperture plane and record the x -component of the electric field in all reconstruction planes and at all frequencies of interest, i.e., $E_x^{\text{inc}}(x, y, z_n, f_l)$, $n = 1, 2, \dots, N_z$ and $l = 1, 2, \dots, N_f$.

One approach to obtain $G_x^x(x, y, z, f_l)$ is to move an x -polarized point source on all reconstruction planes along all pixels and to record the field at the receiving antenna terminals. This approach, however, is extremely inefficient because it requires as many simulations as the number of imaged pixels. Another more efficient approach is using the reciprocity principle [37]. Only one simulation is performed for each transmitting antenna where the infinitesimal x -polarized source is at the origin of the respective aperture plane while the E_x field is sampled simultaneously at all pixels of all reconstruction planes. The collected data from each S -parameter, i.e., $t(x', y', f_l) = s_{jk}(x', y', f_l)$, $j, k = 1, 2$, can be processed separately as described above to create an image of the target. In this case, we can obtain four separate images at each $z = z_n$ plane. Another approach is to reconstruct a single image from the data sets of all S -parameters. To implement this, we write (12) to get N_f equations using the data collected from each S -parameter at each (k_x, k_y) . Considering that $H(k_x, k_y, z_n)$ (or x_n in (12)), $n = 1, 2, \dots, N_z$ (unknowns) are the same in all four systems of equations, these four systems of equations are combined to construct a single system of equations with $4 \times N_f$ rows, each N_f rows correspond to one of the S -parameters. This system of equations is usually over-determined since at each (k_x, k_y) , $4 \times N_f$ equations need to be solved simultaneously to find N_z unknowns $H(k_x, k_y, z_n)$, $n = 1, 2, \dots, N_z$. Again, a least-square solution is sought to find these unknowns.

III. RANGE AND CROSS-RANGE RESOLUTIONS

We approximate the resolution at the center frequency of the system bandwidth $f_c = (f_1 + f_{N_f})/2$. Following the approach presented in [35] for a single-frequency holographic imaging, the cross-range resolution is calculated as [35]

$$\delta_x = \frac{\lambda_c}{2(\sin(\theta_t/2) + \sin(\theta_r/2))} \quad (17)$$

where λ_c is the wavelength corresponding to the frequency f_c , and θ_t and θ_r are the angles subtended by the transmitting and receiving apertures or the full beamwidths of the antennas, whichever is less.

The range resolution δ_z for small targets is approximately [32]

$$\delta_z = \frac{c}{2B}. \quad (18)$$

where c is the wave velocity in the medium and B is the frequency bandwidth.

IV. SPATIAL AND FREQUENCY SAMPLING

In order to make use of all frequency components of the measured signals, the sampling rate of the scan has to satisfy the Nyquist criterion, i.e., the phase shift from one sample point to the next must be less than π . For a spatial sampling interval of Δx along a cross-range direction, the worst case is a phase shift of $2k\Delta x$. Therefore, the sampling criterion can be expressed as [32]

$$\Delta x < \lambda_{\min}/4 \quad (19)$$

where $\lambda_{\min}$ is the wavelength corresponding to the maximum frequency f_{N_f} .

The required frequency sampling is determined in a similar way. The phase shift resulting from a change in the wavenumber Δk is $2\Delta k R_{\max}$, where $R_{\max}$ is the maximum target range. Requiring that this phase shift is less than π yields [32]

$$\Delta f < \frac{c}{4R_{\max}} \quad (20)$$

where Δf is the frequency sampling interval.

V. EXAMPLES AND ARTIFACTS SUPPRESSION

To validate the accuracy of the proposed 3-D microwave holography technique, two $\lambda/2$ (at the center frequency of operation) x -polarized dipole antennas with targets in between are simulated in FEKO Suite 5.4 [38] as shown in Fig. 2. The antennas perform a 2-D scan by moving together on two parallel aperture planes and collecting wideband data at each position. We refer to the apertures as aperture 1 and aperture 2 corresponding to antenna 1 and antenna 2, respectively. The apertures have a size of 60 mm $\times$ 60 mm with their centers being on the z axis.

The S -parameters $s_{jk}^t(j, k = 1, 2)$ for the two antennas are acquired in the presence of the target and recorded for every (x', y') position of the antenna pair. The numerical noise in the acquired data is estimated through the numerical convergence error of the S -parameters which is 0.02.

The proposed holography technique is applied to the calibrated S -parameters. To perform calibration, the same setup is simulated without the target to obtain the background S -parameters $s_{jk}^b(j, k = 1, 2)$. Then, the calibrated S -parameters are calculated as

$$s_{jk}^{\text{cal}}(x', y') = s_{jk}^t(x', y') - s_{jk}^b. \quad (21)$$

Here, due to the uniform background, the background simulations need to be performed only once in a sample (x', y') position since in all other positions they are the same. Since the dipoles are x -polarized, the x -component of the simulated incident electric field E_x^{inc} and the Green function G_x^x are recorded at the reconstruction planes $z = z_n$ as described in Section II. These planes are of size 80 mm $\times$ 80 mm.

To evaluate the quality of the target shape reconstruction process, we define a reconstruction error (RE) parameter as

$$\text{RE}(z_n) = \left\| \frac{|h(x, y, z_n)|}{M} - I(x, y, z_n) \right\| \quad (22)$$

for each slice of the reconstructed image $|h(x, y, z_n)|$, M is the maximum of $|h(x, y, z_n)|$ for all $z_n, n = 1, 2, \dots, N_z$. The function $I(x, y, z_n)$ represents the exact shape of the target. It is 1 inside the target and 0 elsewhere. Also, to have an overall estimate of the reconstruction error, we define the parameter RE^t as

$$\text{RE}^t = \sum_{n=1}^{N_z} \text{RE}(z_n). \quad (23)$$

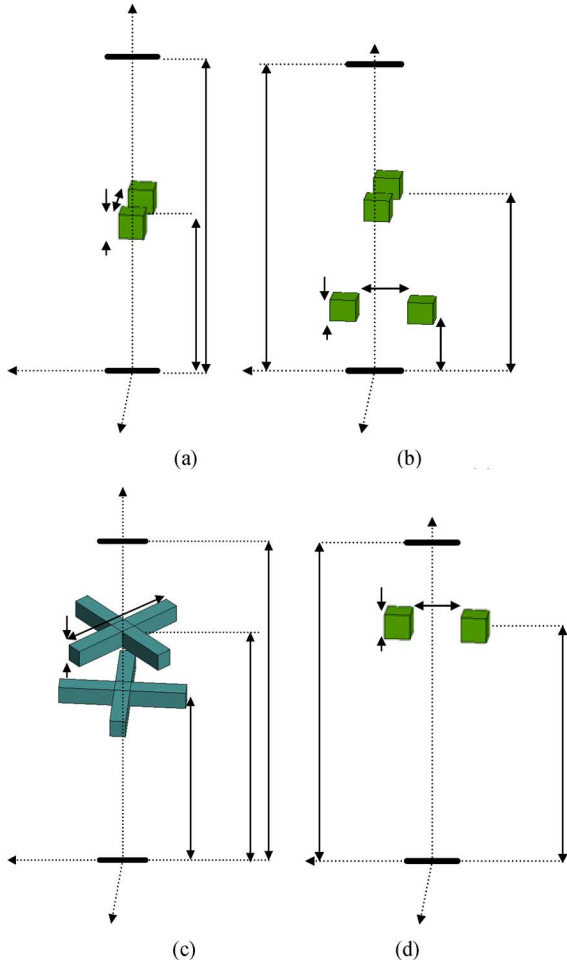


Fig. 2. Dielectric targets in free space scanned by two $\lambda/2$ (at 35 GHz) horizontally-polarized (x -polarized) dipoles; dipole 1 is moving on the $z = 50$ mm plane while dipole 2 is moving on the $z = 0$ mm plane. The simulated S -parameters are recorded in the frequency band of 25 GHz to 45 GHz for: (a) two similar cuboids with sides of 3 mm, centered at $(0, -4.5, 25)$ mm and $(0, 4.5, 25)$ mm with dielectric properties of $\epsilon_r = 2$ and $\sigma = 0$ S/m; (b) four similar cuboids with sides of 3 mm, centered at $(-4.5, 0, 11)$ mm, $(4.5, 0, 11)$ mm, $(0, -4.5, 27)$ mm, and $(0, 4.5, 27)$ mm and all having dielectric properties of $\epsilon_r = 2$ and $\sigma = 0$ S/m; (c) two similar X-shape objects with square cross-sections 2 mm on a side and length of each arm 16 mm, parallel to the x - y plane, one centered at $(0, 0, 27)$ mm with its arms along the x and y axes, the other one centered at $(0, 0, 35)$ mm, with the arms rotated by 45 degrees with respect to the x and y axes, both targets having dielectric properties of $\epsilon_r = 2$ and $\sigma = 0$ S/m; (d) two similar cuboids with sides of d , centered at $(-4.5, 0, 35)$ mm and $(4.5, 0, 35)$ mm, with dielectric properties $\epsilon_r = 5$ and $\sigma = 0$ S/m.

A. Reconstruction Examples for a Background Medium With $\epsilon_r = 1$ and $\sigma = 0$ S/m

In this section, we present a number of reconstruction examples for dielectric targets in air. Fig. 2 shows the simulation setups. In all examples, the aperture planes are located at $z = 50$ mm and $z = 0$. The center frequency is 35 GHz. The wideband data is in the range of 25 GHz to 45 GHz. This gives a range resolution of $\delta_z = 7.5$ mm as per (18). Since $\Delta z \geq \delta_z$ must be fulfilled, in all examples, we set the distance between the reconstruction planes as 8 mm. Since the maximum frequency in the simulations is 45 GHz, as per (19), the spatial sampling rate for the 2-D scan has to be smaller than 1.6 mm. Therefore, a sampling rate of 1.5 mm along x and y is adopted. Also, we approximate the minimum number of frequencies from (20) which

depends on $R_{\max}$. For example, for $R_{\max}$ being 40 mm, the maximum Δf is 1.8 GHz. Here, we use $\Delta f = 2$ GHz which leads to 11 frequencies in the range of 25 GHz to 45 GHz.

In the first example, two dielectric cuboids with a side of 3 mm and wall-to-wall distance of 6 mm are placed midway between the two antennas at $z = 25$ mm as shown in Fig. 2(a). The constitutive parameters of the cuboids are $\epsilon_r = 2$ and $\sigma = 0$ S/m while the background is free space. The separation distance between the reconstruction planes along the range ($z = z_n$ planes, $n = 1, 2, \dots, 5$) is 8 mm. The system of equations in (13) is solved at each spatial frequency pair (k_x, k_y) when the S -parameters are used separately to reconstruct the 2-D images on 5 planes along the z axis. Fig. 3 shows these images. It is observed that the images obtained from s_{11} and s_{22} reconstruct the target at the plane $z = 25$ mm properly but also show some artifacts 8 mm away. The images obtained from s_{21} are not conclusive at all although the image obtained at $z = 25$ mm plane is good. Using s_{12} leads to similar results as s_{21} .

Next, we solve the reconstruction problem when all the S -parameters are used simultaneously. In this case, at each (k_x, k_y) , 44 equations are available to find the 5 unknowns $H(k_x, k_y, z_n)$, $n = 1, 2, \dots, 5$. Fig. 4(a) show the reconstructed images. It is observed that the targets are reconstructed properly at the plane $z = 25$ mm. At the same time, the images at the other range locations do not contain artifacts. The computed values for RE (shown above slice images) and RE^t (shown in the figure captions) also confirm the improvement in the image quality. For this example, the cross-range resolution estimated from (17) is 3.8 mm. Thus, the two targets, which are 6 mm apart, are resolved well.

From now on, we only present the results when the data collected from all S -parameters contribute simultaneously to the image reconstruction.

In the next example, a more challenging reconstruction problem is solved. As shown in Fig. 2(b), a pair of cuboids similar to those in Fig. 2(a) are placed at $z = 11$ mm and another similar pair of cuboids, rotated by 90° about the z axis, are placed at $z = 27$ mm. The constitutive parameters of the cuboids are $\epsilon_r = 2$ and $\sigma = 0$ S/m while the background is free space. Fig. 4(b) shows the reconstructed images. It is observed that the targets are recovered well in the planes at $z = 11$ mm and $z = 27$ mm. Also, the images at the other range locations do not show artifacts.

A third example is shown in Fig. 2(c) where an “X” shape object parallel to the x - y plane is placed at $z = 27$ mm. The “X” shape has two orthogonal arms of length 16 mm and square cross-sections 2 mm on a side. Another similar “X” shape object is rotated by 45° about the z axis and is placed at $z = 35$ mm. The constitutive parameters of the cuboids are $\epsilon_r = 2$ and $\sigma = 0$ S/m while the background is free space. The distance between the two objects is 8 mm, which is very close to the range resolution limit $\delta_z = 7.5$ mm. Fig. 4(c) shows the reconstructed images. The targets are recovered well at $z = 27$ mm and $z = 35$ mm. The images formed at other range locations show weak artifacts. Later, we present an approach to reduce the artifacts. For the target at $z = 27$ mm, the arms along the x axis have been reconstructed better than the arms along the y axis. This is due to the fact that the dipoles are oriented along

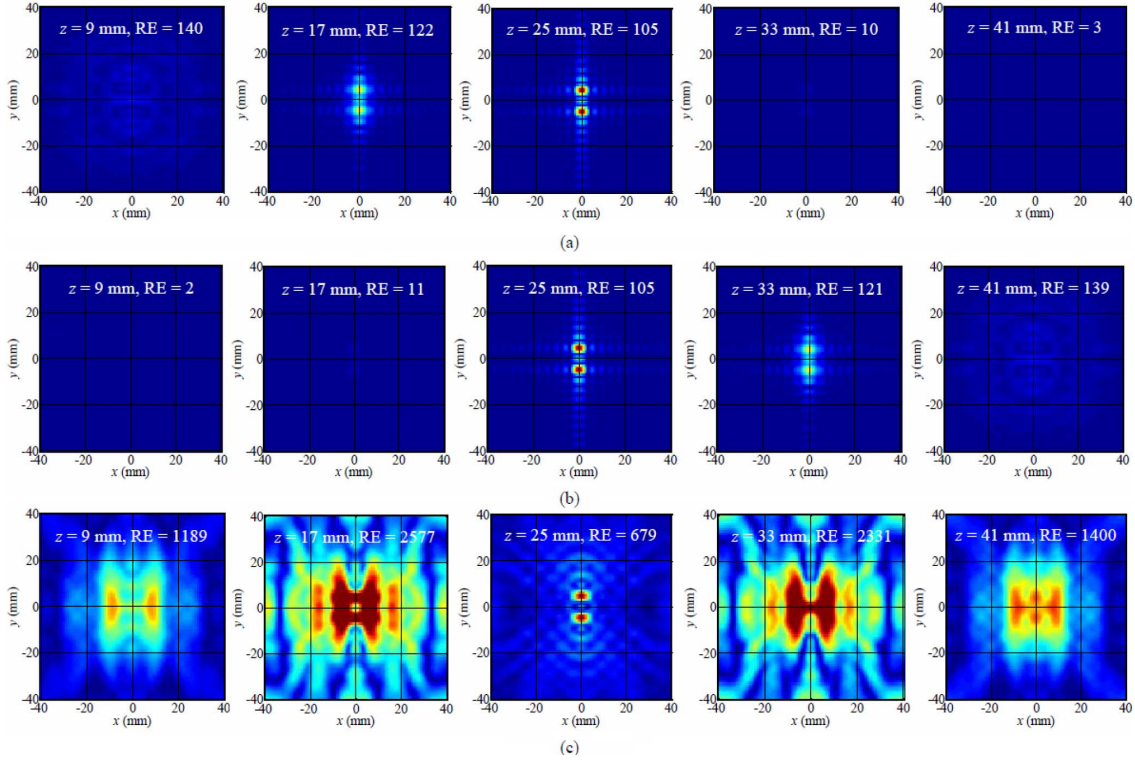


Fig. 3. Reconstructed images of the dielectric targets in Fig. 2(a) when the system of equations in (13) is solved using only: (a) $s_{11}^{\text{cal}}(\text{RE}^t = 380)$, (b) $s_{22}^{\text{cal}}(\text{RE}^t = 378)$, (c) $s_{21}^{\text{cal}}(\text{RE}^t = 8176)$.

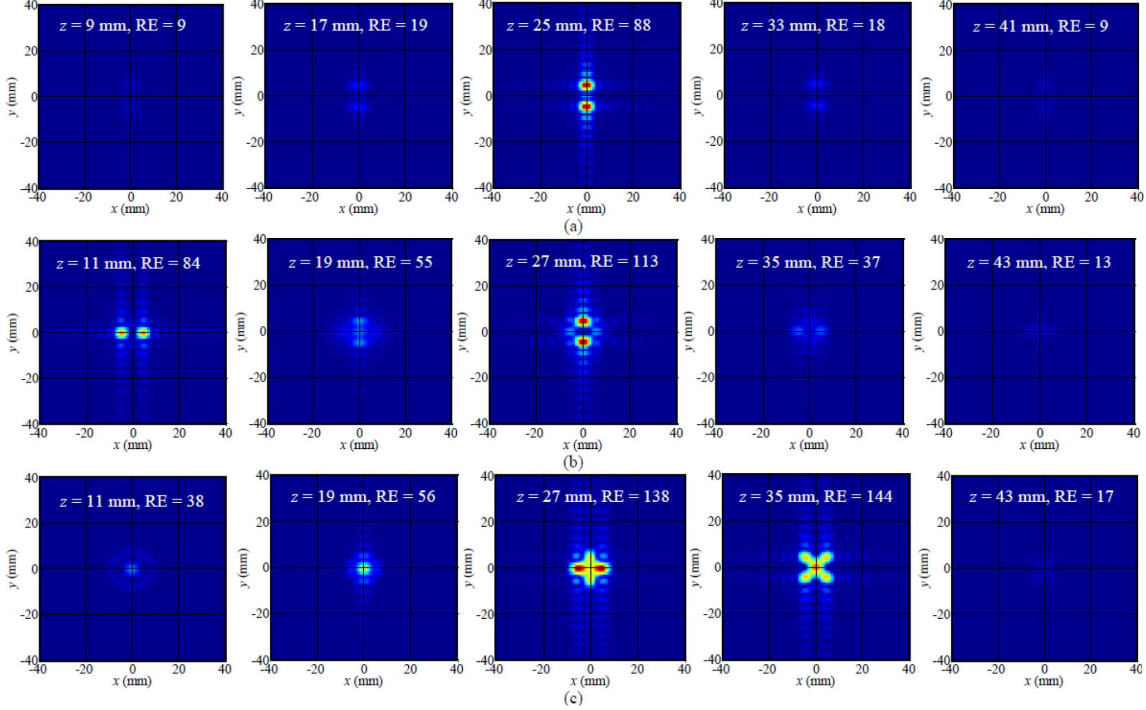


Fig. 4. Reconstructed images when using all S -parameters simultaneously for the dielectric targets in: (a) Fig. 2(a) ($\text{RE}^t = 143$), (b) Fig. 2(b) ($\text{RE}^t = 302$), and (c) Fig. 2(c) ($\text{RE}^t = 393$).

x and thus the incident field is x -polarized. Such field interacts better with the arms oriented along x .

It is worthwhile to compare the proposed technique to the holography technique in [32]. Fig. 5 shows the reconstructed

images for the targets in Fig. 2(c) when applying the 3-D holography technique presented in [32]. Note that this technique makes use of the reflection (back-scattered) signals only. The same number of frequencies is adopted and the reconstructed

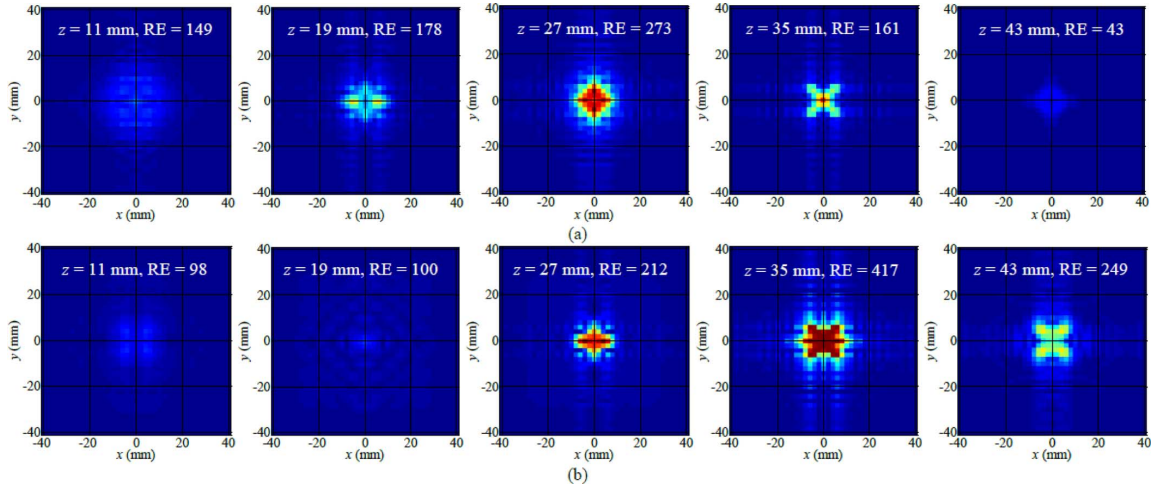


Fig. 5. Reconstructed images obtained for the dielectric targets in Fig. 2(c) when applying the 3-D holography technique in [32] to: (a) $s_{11}^{\text{cal}}(\text{RE}^t = 804)$ and (b) $s_{22}^{\text{cal}}(\text{RE}^t = 1076)$.

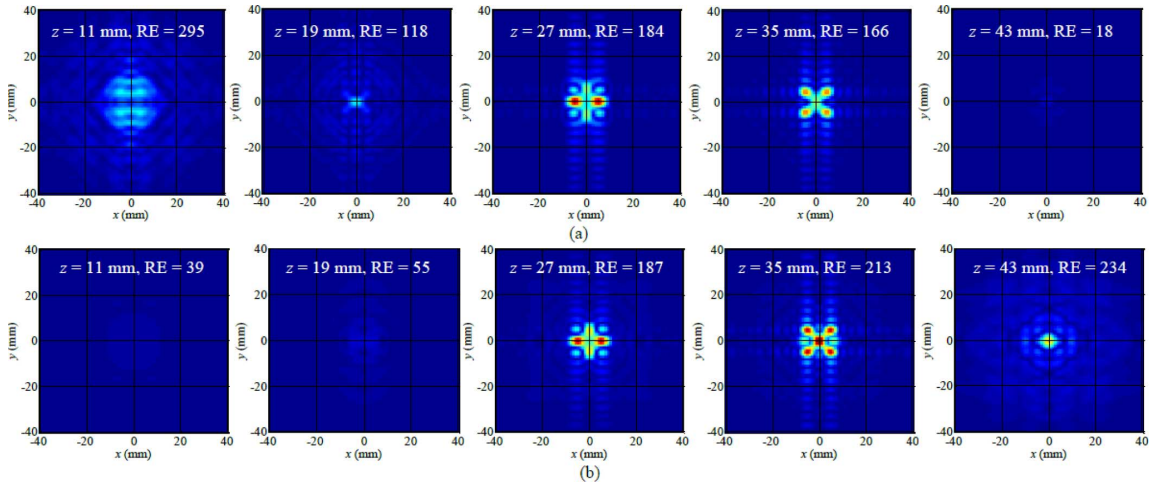


Fig. 6. Reconstructed images obtained for the dielectric targets in Fig. 2(c) when applying our proposed 3-D holography technique to: (a) $s_{11}^{\text{cal}}(\text{RE}^t = 781)$ and (b) $s_{22}^{\text{cal}}(\text{RE}^t = 728)$.

images are shown on the same planes. As observed, artifacts are present and the target shapes are not recovered well. For the sake of comparison, Fig. 6 shows the reconstructed images obtained from our proposed technique where only the reflection S -parameters are processed. The comparison of the RE and RE^t values for the images in Fig. 5 with those in Fig. 6 confirms the advantages of using adequate representation of the incident field and the Green's function in our formulation instead of using the plane-wave assumptions as in [32], which is inadequate in near-field imaging. In addition, the possibility of using transmission S -parameters in conjunction with the reflection ones in our technique leads to even better results as shown in Fig. 4(c). The values of RE and RE^t confirm this observation.

For the three examples presented above, we have used the data recorded at 11 frequencies in the range of 25 GHz to 45 GHz ($\Delta f = 2$ GHz). We observe that when using the data at more frequencies, e.g., 21 frequencies or $\Delta f = 1$ GHz, the quality of the reconstructed images remains almost the same. Also, we observe that very similar results with slight degrada-

tion of quality are obtained when using the simulated data only at 5 frequencies in the same band ($\Delta f = 5$ GHz). However, reducing the number of frequencies to 3 ($\Delta f = 10$ GHz) leads to serious degradation of the reconstructed images. As an example, Fig. 7 shows the results of reconstructing the targets in Fig. 2(b) with the data recorded at 5 and 3 frequencies. As shown in the figure, the RE and RE^t values increase with decreasing the number of sampling frequencies.

B. Reconstruction Example for a Background Medium with $\epsilon_r = 16$ and $\sigma = 0.5$ S/m

Since the late 1970's, when Larsen and Jacobi pioneered the use of scattering parameter imagery for biomedical applications [39], such applications have attracted a lot of attention.

Here, we present an example in which the background medium is lossy dielectric. We chose the dielectric properties and the operating frequency band to be close to those considered in the microwave imaging of biological tissues. In this example, the aperture planes are located at $z = 80$ mm and $z = 0$. As shown in Fig. 8, the targets are X-shape objects cre-

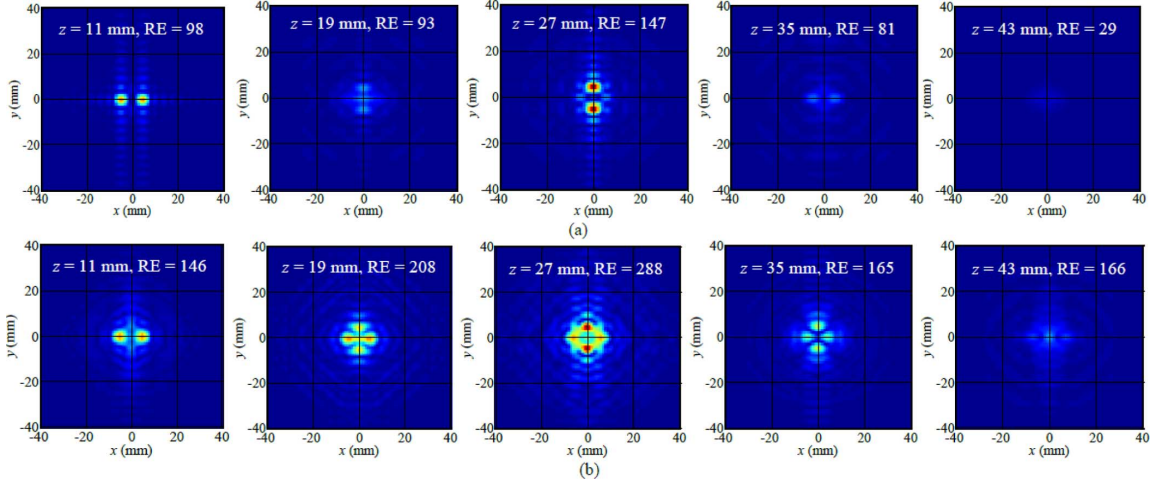


Fig. 7. Reconstructed images obtained for the dielectric target in Fig. 2(b) when using: (a) 5 frequencies ($N_f = 5, \Delta f = 5$ GHz) ($RE^t = 448$) and (b) 3 frequencies ($N_f = 3, \Delta f = 10$ GHz) ($RE^t = 973$).

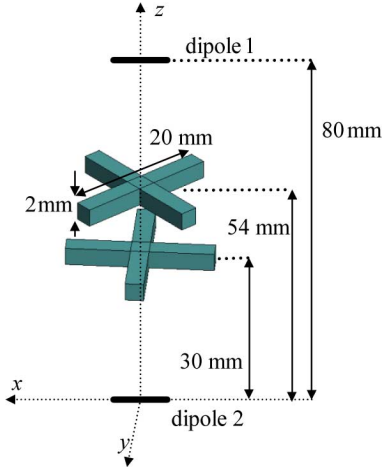


Fig. 8. Dielectric targets with $\epsilon_r = 32$ and $\sigma = 1$ S/m in a background medium with $\epsilon_r = 16$ and $\sigma = 0.5$ S/m scanned by two $\lambda/2$ (at 6.5 GHz) x -polarized dipoles; dipole 1 is moving on the $z = 80$ mm plane while dipole 2 is moving on the $z = 0$ mm plane. The simulated S -parameters are recorded in the frequency band from 3 GHz to 10 GHz for two similar X-shape objects. Each arm of the X-shape has a square cross-section 2 mm on a side and a length of 20 mm. The arms are parallel to the x - y plane. One centered at (0,0,30) mm with its arms along the x and y axes. The other one is centered at (0,0,54) mm, with the arms rotated by 45 degrees with respect to the x and y axes.

ated from two arms of length 20 mm and square cross-sections 2 mm on a side. One target is placed at $z = 30$ mm and the other one, rotated by 45° , is placed at the range position $z = 54$ mm. The constitutive parameters of the targets are $\epsilon_r = 32$ and $\sigma = 1$ S/m while the background is with $\epsilon_r = 16$ and $\sigma = 0.5$ S/m. The S -parameters are collected in the range of 3 GHz to 10 GHz (the center frequency is 6.5 GHz) and this leads to a range resolution limit of $\delta_z = 5.4$ mm, according to (18). Since $\Delta z \geq \delta_z$ must be fulfilled, in all examples, we set the distance between the reconstruction planes as 6 mm. Since the maximum frequency in the simulations is 10 GHz, from (19), the spatial sampling rate for the 2-D scan has to be smaller than 1.9 mm. Therefore, a sampling rate of 1.5 mm along both x and y is adopted. Also, we approximate the minimum number of frequencies from (20) which depends on $R_{\max}$. For example,

for $R_{\max}$ being 70 mm, the maximum Δf is 0.26 GHz. Here, we use $\Delta f = 0.25$ GHz which leads to 29 frequencies in the range of 3 GHz to 10 GHz. Fig. 9 shows the reconstructed images. The targets are recovered well at $z = 30$ mm and $z = 54$ mm. The images formed at other range locations show some artifacts, which are weaker than the images at the slices containing the targets. These artifacts can be reduced further by a method described later.

C. Reconstruction at Higher Contrasts and Larger Sizes

In all examples considered so far, the ratio of the permittivity of the target to that of the background medium was 2. To investigate the quality of the imaging for targets with higher contrasts, a pair of cuboids are placed at $z = 35$ mm as shown in Fig. 2(d). The constitutive parameters of the cuboids in this example are $\epsilon_r = 5$ and $\sigma = 0$ S/m while the background is free space. Three cases are investigated where the cuboids have a side of $d = 2$ mm, 2.6 mm, and 3 mm. Fig. 10(a) shows the reconstructed images for the case of $d = 2$ mm. It is clear that the objects are reconstructed well. However, for larger objects ($d = 2.6$ mm and 3 mm), the quality degrades, i.e., artifacts appear and the RE and RE^t values increase.

The degradation of the images when increasing the size of the cuboids is due to the fact that the criterion for the first-order Born approximation is violated for larger cuboids [40]. This approximation is indeed used in deriving (5) when approximating the field inside the targets with the incident field. The condition for applying first-order Born approximation is that the radius a of a sphere enclosing the target and its refractive index n satisfy [40]

$$(n-1)a < \frac{\lambda_{\min}}{4}. \quad (24)$$

For $\epsilon_r = 5, n = 2.23$. Thus, for $f_{N_f} = 45$ GHz ($\lambda_{\min} = 6.7$ mm), the radius of the object needs to be smaller than 1.4 mm for the linear Born approximation to hold. For the targets with $d = 2.6$ mm and 3 mm, this condition does not hold and thus the quality of the images degrades. We note, however, that the detection capability (or the sensitivity) of the algorithm remains good even at high contrasts.

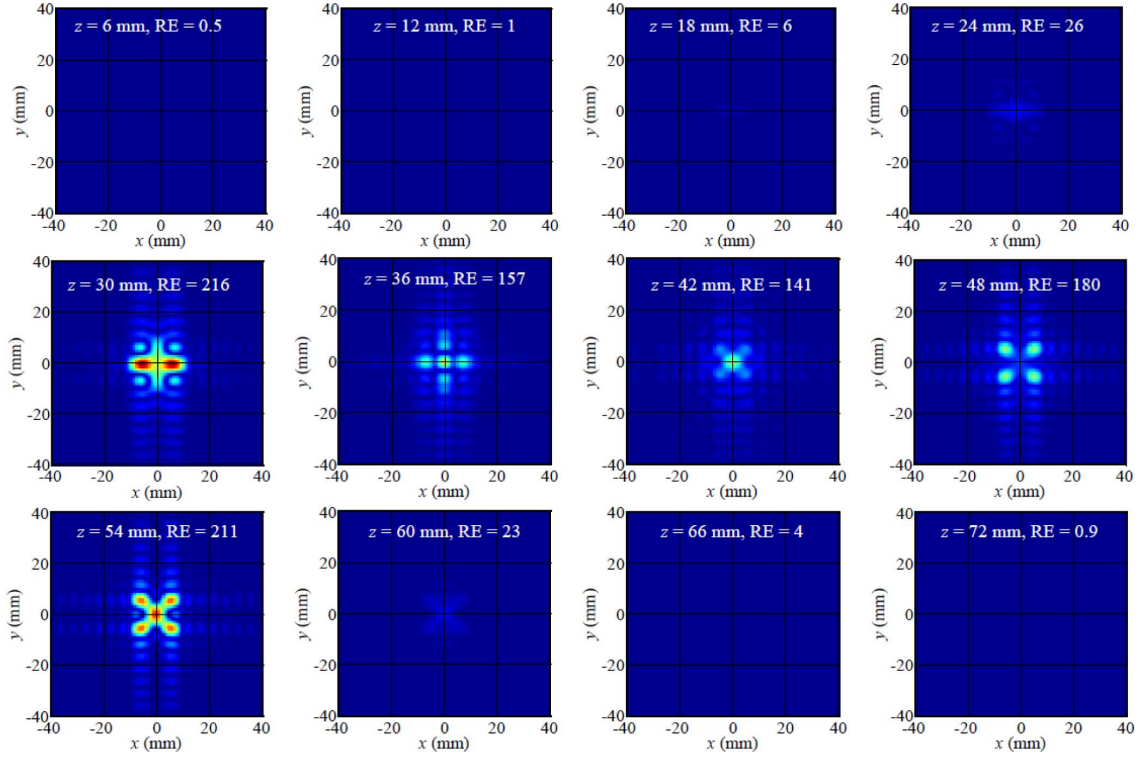


Fig. 9. Reconstructed images for the example shown in Fig. 8 ($RE^t = 966$).

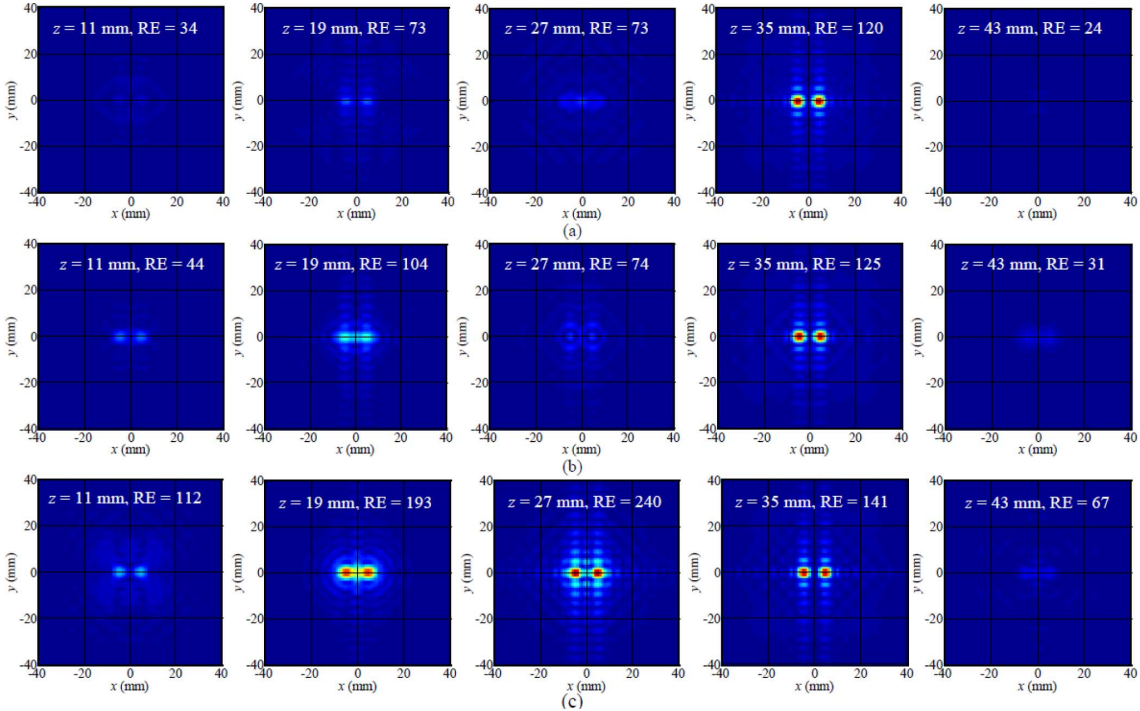


Fig. 10. Reconstructed images obtained from the simulation of dielectric targets in Fig. 2(d) with: (a) $d = 2$ mm ($RE^t = 324$), (b) $d = 2.6$ mm ($RE^t = 378$), and (c) $d = 3$ mm ($RE^t = 753$).

D. Effect of Random Noise on the Reconstruction Results

In order to investigate the effect of random noise on the proposed 3-D holography technique, we consider two types of noise: 1) white Gaussian noise added to the magnitude of the S -parameters (such noise could be produced by electronic

devices or the environment), 2) white Gaussian noise added only to the phase of the S -parameters (such noise could be due to the mechanical vibrations of the antennas). For both types, we apply high levels of noise.

Fig. 11 shows the reconstructed images for the targets in Fig. 2(c) when adding noise of type 1 and 2, respectively. For

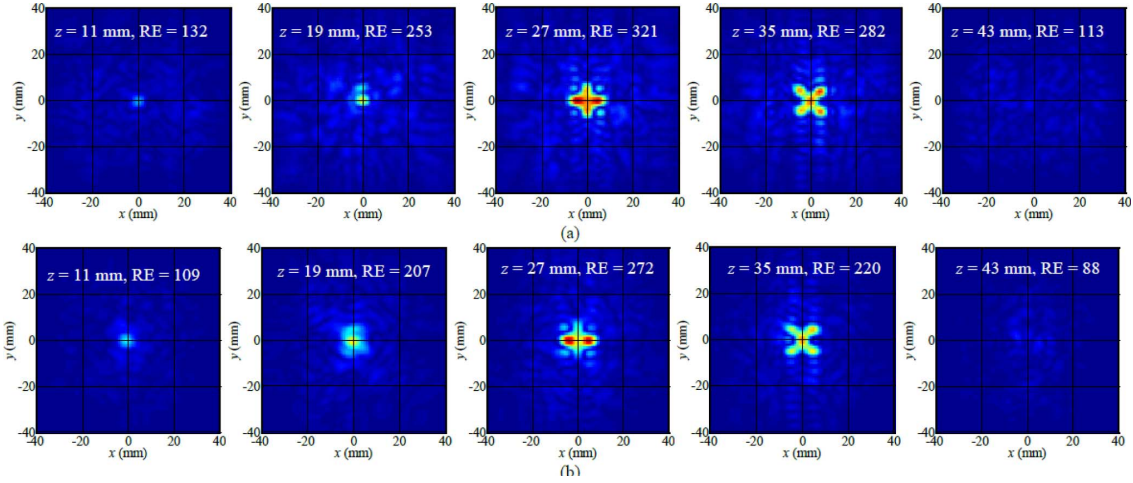


Fig. 11. Reconstructed images for the example of Fig. 2(c) when: (a) adding noise to the complex S -parameters with $\text{SNR} = -10$ dB ($\text{RE}^t = 1101$), and (b) adding noise to only to the phase of the S -parameters with $\text{SNR} = 10$ dB ($\text{RE}^t = 896$).

SNR value as low as -10 dB for the first type of noise and 10 dB for the second type, the quality of the reconstructed images is still comparable to that of the images obtained from noiseless data [see Fig. 4(c)] although RE and RE^t degrade. This indicates that the proposed reconstruction technique is very robust to noise. SNR values lower than -10 dB and 10 dB for noise of type 1 and 2, respectively, lead to visible degradation of the reconstructed images.

E. Suppressing Artifacts Along Range

To improve the quality of the image reconstruction, we make use of a minimization procedure which was first proposed in [35] for the purpose of locating a target along the range in single-frequency 2-D holography imaging. Here, the method is used in the 3-D reconstruction algorithm to suppress artifacts along the range. The process involves the following steps. First, using only the s_{11} data, the images $|h_{11}(x, y, z_n)|$, $n = 1, 2, \dots, N_z$, are obtained. Second, using only the s_{22} data, the images $|h_{22}(x, y, z_n)|$, $n = 1, 2, \dots, N_z$, are obtained. Third, the $|h_{11}(x, y, z_n)|$ and $|h_{22}(x, y, z_n)|$ images are subtracted at each reconstruction plane and the 2-norm of the difference is taken as

$$J(z_n) = \left\| \frac{|h_{11}(x, y, z_n)|}{\max |h_{11}(x, y, z_n)|} - \frac{|h_{22}(x, y, z_n)|}{\max |h_{22}(x, y, z_n)|} \right\|, \quad n = 1, 2, \dots, N_z. \quad (25)$$

Assuming that the artifacts are not similar in $|h_{11}(x, y, z_n)|$ and $|h_{22}(x, y, z_n)|$ unlike the targets, which should appear consistently in both images, we expect that the function in (25) has minima in the true positions of the targets. This can be employed to suppress the artifacts as explained next.

The function $J(z)$ can be used to weigh properly the slice images along the range. Images at range positions where $J(z)$ is small are assigned larger weight than those where $J(z)$ is large. This is implemented using

$$|\hat{h}(x, y, z_n)| = \frac{|h(x, y, z_n)|}{J(z_n)}, \quad n = 1, 2, \dots, N_z \quad (26)$$

where $|\hat{h}(x, y, z_n)|$ is the enhanced image.

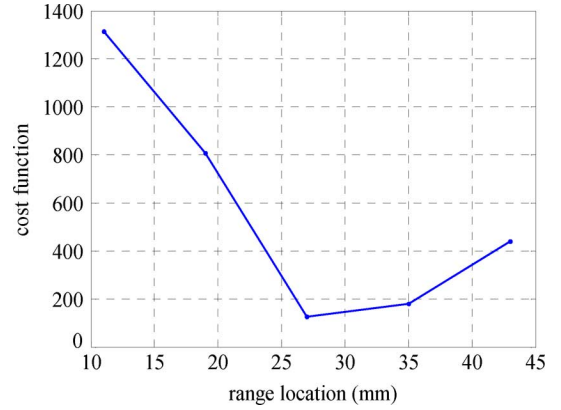


Fig. 12. Variation of cost function in (25) for the example of Fig. 4(c).

As an example, Fig. 12 shows the cost function (25) computed for the example of Fig. 2(c). It has the lowest values at $z = 27$ mm and $z = 35$ mm, where the actual targets are. From the reconstructed images shown in Fig. 4(c), an object is also observed at $z = 19$ mm. However, the large value of the cost function at this range location indicates that this object is an artifact. Fig. 13 compares the enhanced images $|\hat{h}(x, y, z_n)|$ with the original reconstructed images. Note that the RE and RE^t values reduce and the artifact seen at $z = 19$ mm in Fig. 4(c) disappears.

VI. CONCLUSION

A general 3-D holography technique for near-field microwave imaging of dielectric targets is proposed, which allows for incorporating both transmission and reflection data. The data are acquired by two antennas moving on two parallel apertures on the opposite sides of the inspected region and recording wideband data at each position. This technique, compared to previously proposed 3-D holography techniques, has a number of distinct features and advantages. First, the forward-scattered signals contribute to the image reconstruction in addition to the back-scattered signals. This additional information leads to reconstructed images of higher fidelity. It also allows for the efficient suppression of artifacts along range. Second, the incident field is supplied by measurements or simulations and need not be in the ana-

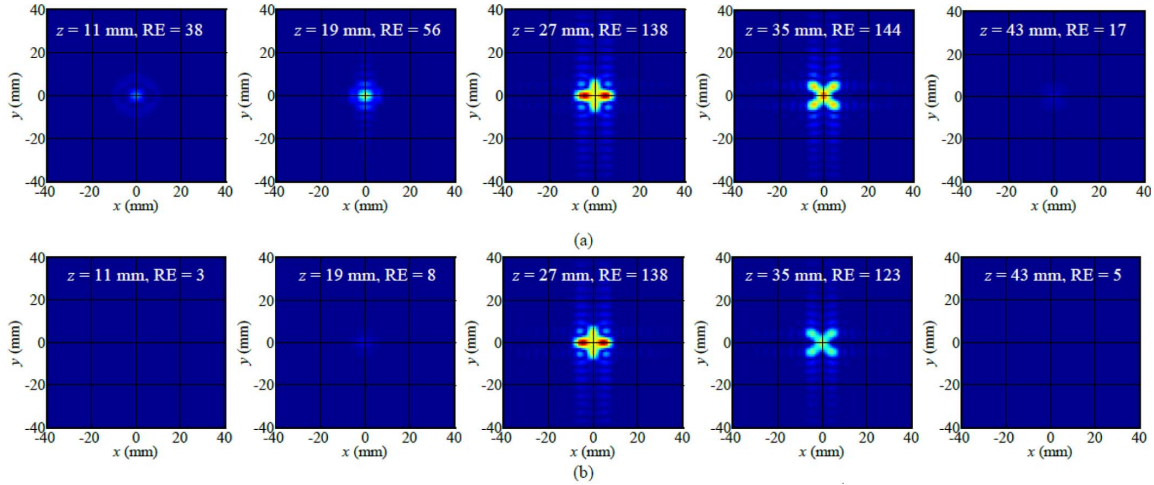


Fig. 13. (a) Original reconstructed images when using all S -parameters simultaneously for the dielectric targets ($RE^t = 393$) and (b) enhanced images using (26) ($RE^t = 277$).

lytical form of plane, cylindrical, or spherical waves. Third, the Green function of the scattering problem is supplied by simulation and need not be approximated by the free-space Green's function. The acquisition of the Green function exploits the reciprocity principle and requires only one simulation. The above two features make this technique suitable for near-field imaging. This advantage is especially important when the target is in the near zone of the antenna because, in this case, the analytical incident-wave representations are inadequate [41]. Fourth, the numerical formats of the incident field and Green's function necessitate a new approach to the solution of the inverse problem, which exploits a least-square solution at each spatial frequency pair (k_x, k_y) . We should emphasize that this approach provides systems of equations with much smaller number of unknowns compared to the systems of equations constructed in regular optimization-based techniques (where the number of unknowns is equal to the number of reconstructed voxels). This reduces the ill-posedness of the proposed technique significantly. Fifth, the algorithm is fast, accurate, and robust to high levels of noise. The execution time is in the order of few minutes on a Pentium 4 with CPU speed of 3 GHz and 4 GB of RAM.

As long as the criterion for the first-order Born approximation is fulfilled, high-quality artifact-free images of the targets are obtained. The technique is capable of providing high-quality 3-D images with the minimum required number of sampling frequencies. It facilitates robust, reliable, and fast near-field microwave imaging of dielectric bodies with possible application in biomedical imaging where the acquisition of both reflected and transmitted signals is feasible. It is worth noting that using cross-polarized data leads to more accurate reconstructed images. This involves using dual-polarization antennas.

APPENDIX

FORMULATION OF INVERSE SCATTERING PROBLEM IN A DIELECTRIC MEDIUM

In a time-harmonic regime and assuming only electric current excitation $\mathbf{J}$ in a nonmagnetic medium, the field is governed by Maxwell's equations

$$\nabla \times \mathbf{E} = -j\omega\mu_0\mathbf{H} \quad (\text{A.1})$$

$$\nabla \times \mathbf{H} = j\omega\mathbf{D} + \mathbf{J} \quad (\text{A.2})$$

where $\mathbf{E}$ and $\mathbf{H}$ are the electric and magnetic field vectors, respectively, and μ_0 and ω are the permeability of the medium and the angular frequency, respectively. From (A.1) and (A.2), the vector Helmholtz equation for the $\mathbf{E}$ -field is obtained:

$$\nabla \times \nabla \times \mathbf{E} - k^2\mathbf{E} = -j\omega\mu_0\mathbf{J} \quad (\text{A.3})$$

where $k^2 = \omega^2\mu_0\varepsilon - j\omega\mu_0\sigma$, and ε and σ are the permittivity and the conductivity of the (isotropic) background medium. Consider two scenarios: 1) the target is not present and 2) the target is present. For these two cases, (A.3) gives

$$\nabla \times \nabla \times \mathbf{E}^{\text{inc}} - k_b^2\mathbf{E}^{\text{inc}} = -j\omega\mu_0\mathbf{J} \quad (\text{A.4})$$

$$\nabla \times \nabla \times \mathbf{E}^{\text{tot}} - k_s^2\mathbf{E}^{\text{tot}} = -j\omega\mu_0\mathbf{J} \quad (\text{A.5})$$

where $\mathbf{E}^{\text{inc}}$ and $\mathbf{E}^{\text{tot}}$ are the incident and total field vectors, respectively, and k_b and k_s are the complex wavenumbers in the investigated region for the cases without and with the target, respectively. From (A.4) and (A.5), we obtain

$$\nabla \times \nabla \times \mathbf{E}^{\text{sc}} - k_b^2\mathbf{E}^{\text{sc}} = -(k_s^2 - k_b^2)\mathbf{E}^{\text{tot}} \quad (\text{A.6})$$

where $\mathbf{E}^{\text{sc}} = \mathbf{E}^{\text{tot}} - \mathbf{E}^{\text{inc}}$. The solution for (A.6) is [23].

$$\mathbf{E}^{\text{sc}}(\mathbf{r}) = \iiint_V \mathbf{G}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{E}^{\text{tot}}(\mathbf{r}') [k_s^2(\mathbf{r}') - k_b^2] d\mathbf{r}' \quad (\text{A.7})$$

where V is the target volume and $\mathbf{G}(\mathbf{r}, \mathbf{r}')$ is the dyadic Green's function. Under certain conditions, $\mathbf{E}^{\text{tot}}$ in (A.7) can be approximated by the incident field, i.e., $\mathbf{E}^{\text{tot}} \approx \mathbf{E}^{\text{inc}}$. This is known as the linear Born approximation for weak or small scatterers [23]. In this case, (A.7) is approximated as

$$\mathbf{E}^{\text{sc}}(\mathbf{r}) = \iiint_V \mathbf{G}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{E}^{\text{inc}}(\mathbf{r}') [k_s^2(\mathbf{r}') - k_b^2] d\mathbf{r}'. \quad (\text{A.8})$$

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